AdifySkills

n8n Automation Mastery
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CLASS NOTES

Module 6 · Production-Grade Engineering

DAY 20 of 25

Scaling, Security & Workflow DevOps

"Ek client se 50 clients tak — bina tootey."

⚡ AAJ HUM YEH BANAYENGE

Nightly Workflow Backup → GitHub

Har raat 2 baje n8n API se saare workflows nikaalta hai aur har ek ko date-wise folder mein GitHub repo mein commit karta hai. Galti se delete ho jaaye to bhi version wapas.

Redis | Queue Mode | GitHub | n8n API

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Ek client se 50 clients tak jaane ke liye scale, security aur version control chahiye — warna ek galti sab tod deti hai.

🎯 Aaj ke baad aap yeh kar paoge

01 Queue mode with Redis & workers
🌍 Order board + kai chef.
⚙️ EXECUTIONS_MODE=queue, Redis, n8n worker containers.

02 Performance: batching, memory, binary data mode
🌍 Ek baar mein 10 plate, 1000 nahi.
⚙️ Split in Batches, binary data mode = filesystem.

03 Security: credentials, RBAC, secrets, webhook auth
🌍 Ghar ki chaabi sabko nahi dete.
⚙️ Credentials store, user roles, webhook header/basic auth.

04 Version control: export JSON, Git, environments
🌍 Recipe book ki photocopy.
⚙️ n8n API → workflows export JSON → GitHub commit.

05 Testing workflows & staging vs production
🌍 Pehle ghar par try, phir party mein.
⚙️ Staging instance → test data → phir production.

06 Monitoring: execution logs, uptime, metrics
🌍 Dukaan ka CCTV.
⚙️ Execution logs, uptime ping, failed executions count alert.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

`0:00-0:10` **Recap**
🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

`0:10-0:45` **Concept: Scale Architecture**
🪄 *Ek chef 100 log ka khana nahi bana sakta. 5 chef aur ek order board chahiye.*

`0:45-1:30` **Live Build**
Workflow backup to GitHub (auto, nightly)
🪄 *Aaj ki kahani yaad hai — 'Restaurant ka bada kitchen'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

`1:30-1:50` **Quiz + Audit**
Module 6 quiz + security audit checklist
🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

`1:50-2:00` **Q&A + Homework**
🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Queue mode

Main instance + Redis + multiple workers — parallel executions.

🪄 **Analogy:** Restaurant: ek order-taker, kai chefs

CONCEPT 02

Binary data mode

Files memory ki jagah disk/S3 par — RAM bachti hai.

🪄 **Analogy:** Saamaan godown mein rakho, kamre mein nahi

CONCEPT 03

Source control

Workflows JSON ko Git mein — history + rollback.

🪄 **Analogy:** Google Docs version history

CONCEPT 04

Least privilege

Har credential ko sirf zaroori permission.

🪄 **Analogy:** Guest ko sirf drawing room ki chaabi

🔑 Aaj ka main idea

Scale = zyada load sambhalna (queue mode, workers, batching). Security = credentials, access aur webhooks ko surakshit rakhna. DevOps = workflows ka version, testing aur monitoring.

- Redis + 3 workers se 10× throughput
- Webhook par header auth
- Har raat workflows GitHub par backup

📖 Ek kahani se samjho: Restaurant ka bada kitchen

1 🧾 **Bahut saare orders aaye**
= Executions queue

2 📋 **Order board par lage**
= Redis queue

3 👨‍🍳 **5 chef saath kaam karein**
= Workers

4 🔐 **Kitchen mein sirf staff**
= RBAC + credentials

5 📸 **Har recipe ki copy**
= Git version control

🪄 Ek line mein: “Ek chef 100 log ka khana nahi bana sakta. 5 chef aur ek order board chahiye.”

🔬 Gehraai se samjho — har concept andar se

Kya hai, n8n andar kya karta hai, kab use karein, kab nahi, example aur common galti.

01 Queue mode

📖 **Kya hai:** Queue mode mein n8n ka ek "main" instance sirf triggers/webhooks sunta hai aur kaam ek queue (Redis) mein daal deta hai; kai "worker" instances us queue se kaam utha kar parallel mein chalate hain.

⚙️ **Andar kaise:** Normal (regular) mode mein ek hi process sab karta hai — bhaari executions aaye to UI aur webhooks dheemi. Queue mode: EXECUTIONS_MODE=queue + Redis + Postgres; workers ko badha kar zyada load sambhalo. Webhook ke liye alag webhook processors bhi laga sakte ho.

✅ **Kab use karein**
Hazaaron executions/ghanta, bhaari AI/file processing, ya bahut saare clients ek hi n8n par.

⛔ **Kab nahi**
Shuru mein (100–1000 executions/din) queue mode ki zaroorat nahi — complexity aur kharcha badhta hai. Pehle regular mode + achha VPS.

🧪 **Example:** 1 main + Redis + Postgres + 3 workers (docker compose scale n8n-worker=3).

⚠️ **Common galti:** Queue mode mein SQLite use karna — chalega hi nahi. Postgres zaroori.

🪄 **Queue = ek order-taker, kai chefs. Load badhe tab.**

02 Binary data mode

📖 **Kya hai:** Files (PDF, images, audio) ko n8n by default memory (RAM) mein rakhta hai. Binary data mode "filesystem" ya "s3" karne se files disk/S3 par jaati hain aur RAM bachti hai.

⚙️ **Andar kaise:** N8N_DEFAULT_BINARY_DATA_MODE=filesystem — har file disk par likhi jaati hai, execution data mein sirf uska reference. Queue mode mein workers alag machines par hon to S3 (ya shared storage) chahiye. Isse "JavaScript heap out of memory" jaise crashes bachte hain.

✅ **Kab use karein**
Jab workflows PDFs, images, audio, bade CSV handle karte hon — Day 16/17 jaise.

⛔ **Kab nahi**
Sirf JSON wale workflows mein farak nahi padta — default theek hai.

🧪 **Example:** .env: N8N_DEFAULT_BINARY_DATA_MODE=filesystem

⚠️ **Common galti:** RAM 2 GB, aur 50 PDFs ek saath memory mode mein — n8n crash, saare running executions fail.

🪄 **Saamaan godown (disk) mein, kamre (RAM) mein nahi.**

03 Source control

📖 **Kya hai:** Workflows ko JSON files ki tarah Git (GitHub) mein rakhna — har badlaav ki history, galti par purana version wapas, team mein review.

⚙️ **Andar kaise:** Do tareeke: (1) Aaj ka build — raat ko n8n API se saare workflows nikaal kar GitHub mein commit (har plan par chalta hai). (2) n8n ka built-in Source Control / Environments (Enterprise) — dev aur prod instances Git branch se sync. Credentials ka secret Git mein nahi jaata, sirf reference.

✅ **Kab use karein**
Client production workflows, team mein kaam, ya jab galti se delete/overwrite ka darr ho.

⛔ **Kab nahi**
Workflow JSON mein agar API keys hardcode ki hain (Code/URL mein) to woh Git mein leak ho jaayengi — pehle unhe credentials mein shift karo. Repo hamesha private.

🧪 **Example:** backups/2026-12-10/day-11-lead-engine-aB12cD.json — har raat ek naya folder.

⚠️ **Common galti:** Public repo mein backup — workflows mein client ke emails, sheet IDs, webhook paths sab duniya ko dikh gaye.

🪄 **Git = version history. Private repo, secrets sirf credentials mein.**

04 Least privilege

📖 **Kya hai:** Har credential, user aur API key ko utni hi permission do jitni kaam ke liye zaroori hai — usse zyada nahi.

⚙️ **Andar kaise:** GitHub token: sirf ek repo, sirf "Contents: read & write". Google: sirf Sheets scope, poora Drive nahi. Supabase: frontend par anon key, n8n par service key. n8n users: client ko sirf apne project ka access. n8n API key: alag key har integration ke liye, taaki ek leak ho to sirf woh revoke karo.

✅ **Kab use karein**
Har naya credential banate waqt — 1 minute ki soch, baad mein bada nuksaan bachata hai.

⛔ **Kab nahi**
"Sab permissions de do, baad mein dekhenge" — baad mein koi nahi dekhta.

🧪 **Example:** Backup workflow ka GitHub fine-grained token: Repository access = only n8n-backups · Contents = Read and write · baaki sab No access.

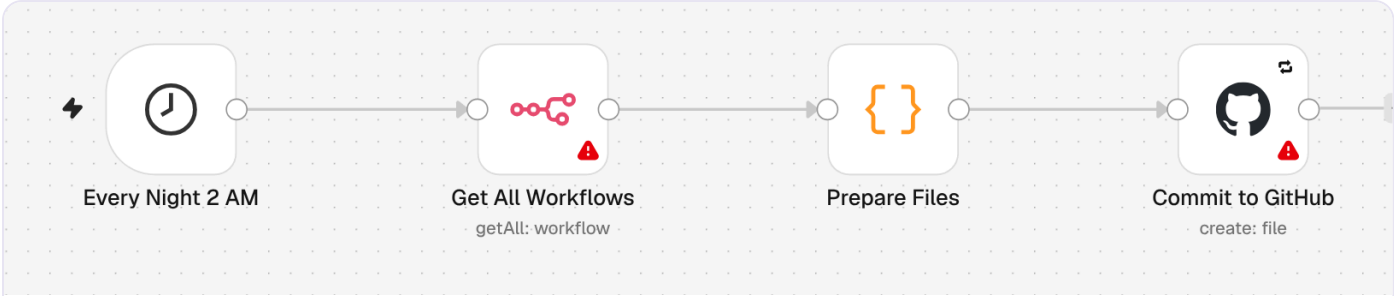
⚠️ **Common galti:** Apna personal GitHub token (saari repos, admin) n8n mein daal dena — n8n compromise = poora GitHub compromise.

🪄 **Guest ko sirf drawing room ki chaabi.**

🗺️ Aaj ka workflow: Nightly Workflow Backup → GitHub

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

Agencies with many workflows (backup + DevOps) — aam taur par **₹5,000 – ₹15,000 add-on**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Schedule Trigger**
Roz raat 2 baje

2 🌐 **n8n (API node)**
Get all workflows

3 🔁 **Loop Over Items (Split in Batches)**
Har workflow ek-ek karke

4 👨‍💻 **Code**
JSON pretty-print, file name = workflow-name.json

5 🐙 **GitHub**
Create/update file in backup repo

6 ✈️ **Telegram**
"✅ 37 workflows backed up"

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **n8n API key (Settings → n8n API)**
- [ ] Connect: **GitHub (personal access token)**
- [ ] GitHub par private repo "n8n-backups" banao
- [ ] n8n Settings → n8n API → API key banao; credential mein Base URL = https://your-n8n-domain/api/v1
- [ ] GitHub token mein "Contents: read & write" permission do

✅ Test kaise karein

1. "Test workflow" → GitHub repo mein backups/<aaj ki date>/ folder check karo

Data ka safar — har node ke baad

Aaj ke build mein data kaise badalta hai. Expressions samajhne ki chaabi yahi hai.

1 🔧 Every Night 2 AM

Raat 2 baje trigger.

```
{
  "timestamp": "2026-12-10T02:00:00.010+05:30"
}
```

2 🧩 Get All Workflows

Har workflow ek item — poori JSON (nodes, connections, settings).

```
{
  "id": "aB12cD",
  "name": "Day 11 — Lead Engine",
  "active": true,
  "nodes": [
    "(9 nodes)"
  ],
  "connections": {},
  "updatedAt": "2026-12-09T18:22:41.000Z",
  "…": "(+36 aur workflows)"
}
```

3 📄 Prepare Files

Har workflow ka path, content, message.

```
{
  "path": "backups/2026-12-10/day-11-lead-engine-aB12cD.json",
  "content": "{\n  \"id\": \"aB12cD\",\n  \"name\": \"Day 11 — Lead Engine\",\n  …",
  "message": "Backup Day 11 — Lead Engine (2026-12-10)",
  "…": "(+36 aur items)"
}
```

4 🐙 Commit to GitHub

Har file ka commit ho gaya.

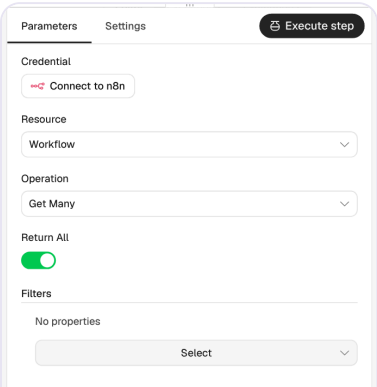
```
{
  "content": {
    "name": "day-11-lead-engine-aB12cD.json",
    "path": "backups/2026-12-10/day-11-lead-engine-aB12cD.json",
    "sha": "3f9a1c…"
  },
  "commit": {
    "message": "Backup Day 11 — Lead Engine (2026-12-10)"
  }
}
```

🧩 Node anatomy — har setting ka matlab

Har zaroori option kya karta hai aur kab badalna hai. Isse aap khud naye workflows bana paoge.

🌐 n8n node (API)

Apne hi n8n se saare workflows ki JSON nikaalta hai.

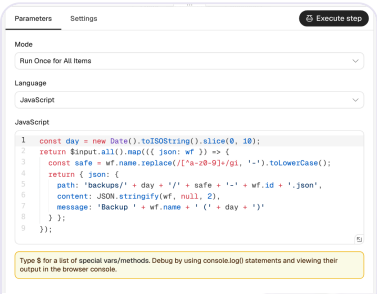


📸 Asli n8n: n8n node (API)

Setting	Kya karti hai	Aaj ki value	Kab badlein
Credential	n8n API key + Base URL	`https://your-n8n-domain/api/v1`	Settings → n8n API → Create an API key
Resource / Operation	Workflow → Get Many	`getAll`	Credentials, executions, tags bhi la sakte ho
Return All	Saare workflows	`On`	Off par sirf 100
Filters	Active only, tags	—	Sirf "production" tag wale backup karne ho to

👨‍💻 Code — file path + content

Har workflow ko ek file ka path, content aur commit message banata hai.

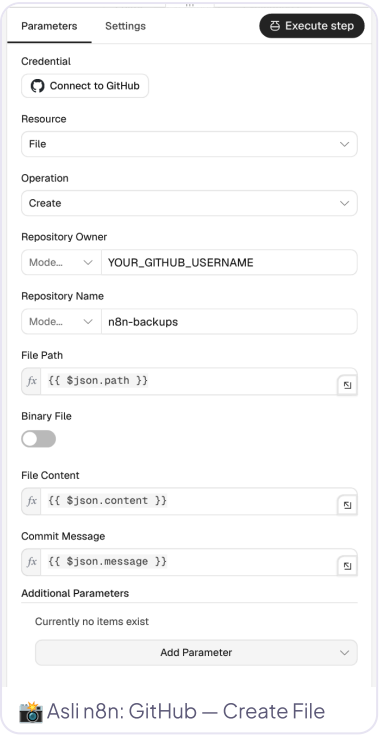


📸 Asli n8n: Code — file path + content

Setting	Kya karti hai	Aaj ki value	Kab badlein
day	Aaj ki date folder ke liye	`new Date().toISOString().slice(0, 10)`	2026-12-10
safe name	Naam se file-safe slug	`wf.name.replace(/[^a-z0-9]+/gi, "-").toLowerCase()`	Spaces, emoji, — sab hata kar -
path	backups/<date>/<slug>-<id>.json	—	ID jodne se same naam wale 2 workflows clash nahi karte
content	Pretty JSON	`JSON.stringify(wf, null, 2)`	Git diff padhne layak

GitHub — Create File

Har file ko repo mein commit karta hai.



📸 Asli n8n: GitHub — Create File

Setting	Kya karti hai	Aaj ki value	Kab badlein
Credential	Fine-grained token	`GitHub account`	Sirf n8n-backups repo, Contents read & write
Resource / Operation	File → Create	`create`	Same path dobara ho to Edit chahiye
Repository Owner / Name	Kahan	`YOUR_GITHUB_USERNAME / n8n-backups`	Repo PRIVATE
File Path / Content	Code node se	`{{ $json.path }} / {{ $json.content }}`	—
Commit Message	Har commit ka message	`Backup Day 11 — Lead Engine (2026-12-10)`	History mein dhoondhna aasaan
Settings → Retry On Fail	GitHub rate limit par	`On`	40+ workflows ho to Loop + Wait bhi socho

Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

Ek server, ek line mein sab — slow

→

⚡ Ab (n8n)

Queue mode: workers parallel

⏱ 1000 webhooks/minute bhi bina atke.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
EXECUTIONS_MODE=queue
QUEUE_BULL_REDIS_HOST=redis
N8N_DEFAULT_BINARY_DATA_MODE=filesystem
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "workflows": 37
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "committed": 37,
  "repo": "yourbrand/n8n-backup"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Har raat 2 baje n8n API se saare workflows nikaalta hai aur har ek ko date-wise folder mein GitHub repo mein commit karta hai. Galti se delete ho jaaye to bhi version wapas.

Flow: Schedule (2:00 AM) → n8n API (Saare workflows) → Code (File path + JSON) → GitHub (Commit file)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt — 📋 Select karke copy karo

```
I'm building the n8n workflow "Nightly Workflow Backup → GitHub" (nodes: Every Night 2 AM, Get All Workflows, Prepare Files, Commit to GitHub).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
4 nodes · ready to import
`adifyskills.com/workflows/day-20-nightly-github-backup.json`

🌐 Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/20`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "n8n API key (Settings → n8n API)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "GitHub (personal access token)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.

⚠️ Yeh galtiyan sabse common hain

- ! Production mein direct edit — staging nahi
- ! API keys workflow JSON mein hardcoded (Git par leak)
- ! Execution data kabhi prune na karna — DB full

⭐ Pro tips — time bachane wali tricks

- ★ EXECUTIONS_DATA_PRUNE=true + max age set karo
- ★ Credentials kabhi export mat karo Git mein
- ★ Uptime monitoring (UptimeRobot) on /healthz

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Nightly workflow: saare workflows export karke GitHub repo mein commit karna.

📝 Homework

Apne instance ka security audit checklist ke hisaab se karo aur report submit karo.

- ☐ Homework ho gaya

BEGINNER
Workflow manually export/import ☐

INTERMEDIATE
Nightly GitHub backup ☐

ADVANCED
Staging → production promotion flow ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Backup ke baad Telegram par "✅ 12 workflows backed up" message aaye.

💡 Hint 1
Commit ke baad saare items ko gino.

💡 Hint 2
Aggregate ya Code node: return [{json: {count: $input.all().length}}]

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🧪 Expression practice sheet

Input dekho aur khaali jagah mein expression likho. Jawab aakhri page par hain.

1. Workflow naam se file slug

```
{"name":"Day 11 — Lead Engine"}
```

Mera jawab: _____

2. File path banao

```
{"id":"aB12cD"}
```

Mera jawab: _____

3. Workflow mein kitne nodes?

```
{"nodes":[{},{},{}]}
```

Mera jawab: _____

4. Active ho to 🟢

```
{"active":true}
```

Mera jawab: _____

5. Pretty JSON kitni lines?

```
{"a":1}
```

Mera jawab: _____

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Parallel executions ke liye mode?

- (A) Queue mode
- (B) Manual mode
- (C) Test mode
- (D) Pin mode

Q2. Queue mode ko kya chahiye?

- (A) SMTP
- (B) Redis
- (C) MongoDB
- (D) FTP

Q3. Workflows ki history ke liye?

- (A) Email
- (B) Pen drive
- (C) Git / source control
- (D) Screenshots

Q4. DB full hone se bachne ke liye?

- (A) Bada font
- (B) More nodes
- (C) Kuch nahi
- (D) Execution pruning

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

API
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

Credential
Kisi app ka login/API key jo n8n safe tareeke se yaad rakhta hai.

Execution
Workflow ka ek baar chalna. Har run ka record Executions tab mein dikhta hai.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 21

Project Sprint 1: AI Customer Support Desk

"24×7 support jo kabhi chhutti nahi leta."

☐ **Day 20 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → A
Queue mode

Q2 → B
Redis

Q3 → C
Git / source control

Q4 → D
Execution pruning

✅ Challenge ka solution

Commit to GitHub → Code: return [{json: {count: $input.all().length}}] → Telegram "✅ {{ $json.count }} workflows backed up".

🧪 Expression sheet ke jawab

1. `{{ $json.name.replace(/[^a-z0-9]+/gi, "-").toLowerCase() }}` → **day-11-lead-engine**
2. `{{ "backups/2026-12-10/" + $json.id + ".json" }}` → **backups/2026-12-10/aB12cD.json**
3. `{{ $json.nodes.length }}` → **3**
4. `{{ $json.active ? "🟢" : "⚪" }}` → 🟢
5. `{{ JSON.stringify($json, null, 2).split("\n").length }}` → **3**

📚 Official docs

- Queue mode ↗
- Security ↗
- Source control ↗
- n8n public API ↗

▶ Videos

- n8n queue mode workers redis ↗
- n8n backup workflows github ↗

Doubt poochho
WhatsApp: +91 83838 91716
wa.me/918383891716

Day 20 lesson page
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